90-Day Personal Learning & Engineering Roadmap (Months 1–3)

Roadmap Attribute	Specification
Engineer	**Shri Sanjaykumar V** (M.Tech Integrated in Software Engineering, VIT Vellore — CGPA: 9.12 / 10.0)
Baseline Project	**FinDoc-AuditEngine v3.0** (Working FastAPI Prototype — Day 29: `28 passed`, Day 30: `33 passed`)
Primary Objective	Transition prototype skills into production-grade persistence, calibrated ML evaluation, MLOps, and distributed backend engineering

Month 1 (Weeks 1–4 / Days 1–30): Durable Persistence, Security & Optical Document Extraction

Goal: Address the core prototype limitations of `FinDoc-AuditEngine v3.0` *by replacing in-memory state, adding authentication, and integrating true optical OCR.*

Week	Focus Domain	Concrete Engineering Tasks	Measurable Weekly Deliverable
Week 1 (Days 1–7)	**Relational Persistence & Async ORM**	Replace the in-memory `AUDIT_LEDGER` Python list with **PostgreSQL** using `SQLAlchemy 2.0` (`asyncpg`) and **Alembic** schema migrations. Design normalized tables for `vendors`, `purchase_orders`, `grn_records`, `invoices`, and `audit_decisions`.	Containerized FastAPI + PostgreSQL service where audit records persist across server restarts.
Week 2 (Days 8–14)	**Idempotency, Concurrency & Caching**	Integrate **Redis** to enforce idempotency keys (`X-Idempotency-Key` with 24-hour TTL) so duplicate webhook submissions cannot create duplicate ledger entries, plus sliding-window vendor velocity counters.	Automated concurrency test suite verifying zero duplicate ledger writes under parallel requests.
Week 3 (Days 15–21)	**Authentication, RBAC & Cryptographic Integrity**	Implement **OAuth2 + JWT** role-based access control (`Auditor`, `Manager`, `Director`, `Admin` scopes) and SHA-256 hash-chaining (`prev_hash -> record_hash`) across PostgreSQL audit rows.	Protected API routes with role-based integration tests and tamper-evident ledger verification script.
Week 4 (Days 22–30)	**Optical OCR & Layout-Aware PDF Parsing**	Upgrade `/api/v1/audit/parse-raw` from regex text parsing to accept real PDF/image invoices using **Tesseract OCR** and layout-aware table extraction (`pdfplumber` / `LayoutLMv3`), plus context-aware NER (`Microsoft Presidio`).	End-to-end PDF upload endpoint extracting multi-line SKU tables with measured field-level F1 accuracy.

Month 2 (Weeks 5–8 / Days 31–60): Applied ML Evaluation, Calibration, MLOps & CI Quality Gates

Goal: Move beyond synthetic baselines to rigorous model evaluation, probability calibration, drift monitoring, and reproducible container benchmarks.

Week	Focus Domain	Concrete Engineering Tasks	Measurable Weekly Deliverable
Week 5 (Days 31–37)	**Benchmarking on Public Financial Datasets**	Evaluate `IsolationForest`, `LocalOutlierFactor`, and supervised `XGBoost` / `LightGBM` on public tabular anomaly datasets (e.g., IEEE-CIS / Credit Card Fraud) using stratified cross-validation.	Reproducible evaluation notebook & report comparing ROC-AUC, PR-AUC, and Recall@K at fixed false-positive rates.
Week 6 (Days 38–44)	**Probability Calibration & SHAP Verification**	Apply **Platt Scaling** and **Isotonic Regression** (`CalibratedClassifierCV`) to convert raw anomaly scores into calibrated probabilities (evaluated via Brier score & reliability diagrams), and compare local perturbation against `TreeSHAP`.	Calibration module with reliability plots and quantitative fidelity comparison between Perturbation XAI and SHAP.
Week 7 (Days 45–52)	**MLflow Registry & Concept Drift Telemetry**	Track model hyperparameters, scaler artifacts, and threshold configs in **MLflow**. Implement weekly **Population Stability Index (PSI)** drift detection across all 7 features using **EvidentlyAI**.	Automated drift-check CLI that flags `PSI > 0.20` and triggers a candidate retraining workflow.
Week 8 (Days 53–60)	**Code Coverage Gating & Load Benchmarking**	Add `pytest-cov` (`>= 85%` line & branch coverage gate) to GitHub Actions CI, export models to **ONNX Runtime**, and run repeatable **Locust** load benchmarks documenting hardware specs and p50/p95/p99 latency.	Published benchmark report with verified p95 latency and CI badge enforcing test + coverage gates.

Month 3 (Weeks 9–12 / Days 61–90): Hybrid RAG, Stateful Agent Workflows & System Design

Goal: Combine structured financial auditing with grounded document retrieval, stateful human-in-the-loop agents, and interview-ready system design.

Week	Focus Domain	Concrete Engineering Tasks	Measurable Weekly Deliverable
Week 9 (Days 61–67)	**Hybrid BM25 + Dense RAG & Reranking**	Build a procurement contract retrieval engine combining sparse **BM25** and dense vector embeddings (`bge-small-en`) via **Reciprocal Rank Fusion (RRF)** and cross-encoder reranking.	Contract QA API citing exact vendor Master Service Agreement (MSA) clauses alongside 3-way match results.
Week 10 (Days 68–75)	**Automated RAG Evaluation & Guardrails**	Integrate **RAGAS** / **DeepEval** to measure Context Precision, Faithfulness, and Answer Relevance on a 50-question procurement test set, plus input prompt-injection filters.	Automated evaluation pipeline verifying Faithfulness `>= 0.90` and zero ungrounded numerical claims.
Week 11 (Days 76–82)	**Stateful LangGraph Dispute Workflows**	Build a cyclic **LangGraph** workflow with persistent Postgres checkpointing (`PostgresSaver`) that drafts vendor discrepancy notices and pauses for human Manager/Director approval before dispatch.	Multi-turn human-in-the-loop agent demo with state resumption and audit logging.
Week 12 (Days 83–90)	**Async Task Queues & Interview Readiness**	Decouple heavy PDF/ML batch processing using **Celery + Redis / Kafka**, complete 60+ medium/hard DSA problems, and write 4 distributed AI/FinTech system design case studies.	Finalized public GitHub portfolio, system design handbook, and mock technical interview completion.